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A Chronic Kidney Disease Diagnostic Model Based on An Interpretable Deep Belief Rule Base

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ABSTRACT Chronic Kidney Disease (CKD) has become a serious public health problem because of its characteristic 'three highs and one low': high prevalence, high disability rate, high medical costs, and low awareness. Therefore, an accurate diagnosis of CKD is crucial. Given the unique nature of the medical field, it is essential to ensure that the results of the CKD diagnostic model are trustworthy to doctors and patients and are practically applicable. Belief Rule Base (BRB) models explain their results through transparent reasoning processes and belief distributions. However, existing BRB models are typically constructed as one-time setups that require reconstruction to include new patient indicators. Furthermore, redundant rules within BRB models increase the difficulty of optimization, and randomness in the optimization process can affect the interpretability of the model. To address these challenges, this paper presents a CKD diagnostic model based on an interpretable Deep Belief Rule Base (DBRB-I). The model leverages a deep BRB structure to enhance scalability and prevent rule explosions. Additionally, it introduces a novel rule reduction method that incorporates expert knowledge to simplify redundant rules in an interpretable way, thus easing the optimization process. To preserve the model's interpretability, the optimization algorithm includes constraints grounded in expert knowledge. Case studies and comparisons with seven other methods show that the CKD diagnostic model based on DBRB-I outperforms the others in a practical diagnostic setting.

INDEX TERMS Belief rule base, disease diagnosis, rule reduction, evidence reasoning

I. INTRODUCTION

THE kidneys regulate homeostasis of blood components, allowing them to eliminate waste, maintain electrolyte balance, control overall fluid balance, and produce hormones [1]. Chronic Kidney Disease (CKD) is a general term that encompasses various symptoms of decreased kidney function lasting more than 3 months [2]. In the early stages, patients with CKD may have no obvious symptoms or only nonspecific symptoms such as drowsiness, itching, or loss of appetite [3]. CKD is an irreversible progressive disease that leads to high rates of morbidity and mortality among patients [4]. According to statistics, at least 2.4 million people worldwide die of CKD each year and it is expected that by 2040, CKD will become the fifth leading cause of death globally [5]. Once patients reach kidney failure, they must rely on dialysis or kidney transplantation to maintain their lives [6]. Dialysis necessitates regular hospital visits, which result in significant inconvenience to patients' daily lives and work. Due to limitations in kidney availability and patients' health conditions, only a limited number of individuals can successfully undergo kidney transplantation. Early detection and intervention of CKD can significantly reduce complications and prolong patients' lifespans, making accurate diagnosis essential.

In recent years, the continuous advancement of artificial intelligence technology has led to an increasing application of machine learning in the diagnosis of CKD. Numerous researchers have made substantial contributions to the development of diagnostic models for CKD. Models are primarily categorized into three types based on the user's understanding of the internal structure and implementation details: black-box, white-box, and gray-box models.

Black-box models are widely favored in practical applications because of their high predictive capability. Their strong performance comes from their reliance on input information to set parameters and construct the model. For example, Singh et al. [7] proposed a novel deep-learning model for the early detection and prediction of CKD. Recursive Feature Elimination (RFE) was employed to select the most important features. The proposed deep neural network model outperformed four other classifiers (SVM, KNN, logistic regression, random forest, and naive Bayes classifiers) achieving 100% precision in classification. Islam et al. [8] explored the potential of various machine-learning algorithms for the early diagnosis of CKD. Among 12 different machine learning classifiers tested, the XgBoost classifier performed the best with an accuracy of 0.983, and precision, recall, and F1 scores of 0.98. Jongbo et al. [9] developed an ensemble method for the diagnosis of CKD. Their study demonstrated that when using the K Nearest Neighbors (KNN) classifier, the random subspace ensemble achieved a prediction accuracy of 100%. Polat et al. [10] used the Support Vector Machine (SVM) classification algorithm combined with feature selection methods to diagnose CKD. The results indicated that using the filter subset evaluator with the best first search algorithm (Best First search engine) for feature selection, the SVM classifier achieved higher accuracy (98.5%) in diagnosing CKD compared to other selected methods. While black-box models excel in predictive performance, they remain completely opaque regarding feature transformation and decision-making processes. In fields such as CKD diagnosis, where transparency and interpretability are essential, results from black-box models may not align well with expert knowledge. This lack of transparency can undermine the reliability and trustworthiness of decisions made by these models.

In contrast to black-box models, white-box models integrate medical principles and expert knowledge during their construction, particularly in the definition of features and interpretation of outputs. The internal mechanisms of white-box models are more transparent and easier to understand, often offering detailed explanations of the decision-making process. For example, Chaudhuri et al. [11] proposed a novel approach for diagnosing CKD by employing Recursive Feature Elimination (RFE) to select optimal feature subsets, combined with the Enhanced Decision Tree (EDT) algorithm. Their study demonstrated that even after removing less important features, the accuracy of EDT remained unchanged. Chaurasia et al. [12] proposed a decision tree-based data mining approach for diagnosing CKD, utilizing both regression and classification tree algorithms to process and analyze data. Through cross-validation, the model achieved approximately 93% accuracy on the test set, indicating high precision. Pasadana et al. [13] explored the diagnosis of CKD using various decision tree techniques and compared their performance. Ultimately, the results showed that Random Forest achieved the highest accuracy of 100% in identifying CKD. Ripon et al. [14] initially derived rules for the relationship between renal function indicators using two rule extraction techniques: Ant-Miner and decision trees. They then used two booster classifiers to diagnose CKD. Although white-box models offer transparent and interpretable decision-making processes, their reliance on medical principles and expert knowledge may limit their predictive accuracy. This may result in less precise outcomes compared to more data-driven black-box models.

Gray-box models aim to combine the strengths of both black-box and white-box models, providing a relatively balanced solution. These models offer high prediction accuracy while maintaining a degree of transparency. By incorporating interpretable features and rules alongside complex algorithms, gray-box models enhance predictive performance, meeting both accuracy and transparency requirements in medical fields, particularly in CKD diagnosis. For example, Qin et al. [15] proposed a machine learning approach to diagnosing CKD, using KNN for data imputation. Among six algorithms, random forest was most accurate at 99.75%, while an integrated model of logistic regression and random forest achieved 99.83% accuracy, showing promise for complex clinical data. Rashed et al. [16] developed machine learning models to improve the early diagnosis of chronic kidney disease (CKD) using key clinical test attributes. Optimized datasets with these attributes achieved high diagnostic accuracy, with the random forest classifier performing the best. This approach offers a cost-effective method for early CKD screening and better patient management. Zhao et al. [17] developed and validated a CKD diagnostic model based on random forest regression, predicting the results of the electronic medical record (EMR) extracted from the data of a regional health system. They built the model using random forest regression and evaluated its performance using goodness of fit statistics and discriminative metrics. Samet et al. [18] developed a machine-learning model to predict CKD using the CKD database. By combining imputation methods and feature selection, the optimized Random Forest method achieved 99. 24% precision, offering strong support for the diagnosis and prevention of CKD.

BRB, a typical gray-box model that employs a hybrid intelligent human-machine modeling approach proposed by Yang et al. [19], is known for its reasonable model parameters, transparent reasoning process, and traceable output, which makes it safe and reliable. The BRB model consists of a series of rules defined by experts, indicating its ability to handle fuzzy and uncertain information effectively under expert guidance, thus aiding in making informed decisions. To date, BRB has shown excellent performance in various fields such as safety assessment [20]–[22], fault diagnosis [23]–[25], and intelligent decision-making [26]–[28].

BRB has also proven to be effective in the diagnosis of various diseases in the medical field. For example, Zhou et al. [29] successfully diagnosed gastric cancer lymph node metastasis using a clinical decision support system based on the Cooperative Belief Rule Base (CBRB). The CBRB consists of two independent BRBs, each handling tumor and lymph node characteristics, and ultimately combines output through an Evidence Reasoning (ER) method. The study indicated that CBRB outperformed commonly used Support Vector Machine (SVM) and Artificial Neural Network (ANN) in diagnosing gastric cancer lymph node metastases while embedding more expert knowledge. Kong et al. [30] proposed a belief rule base inference methodology to assess the clinical risks of chest pain related to the heart. The research demonstrated that in the presence of clinical uncertainty, this method offers more reliable and informative diagnostic suggestions compared to traditional human diagnoses. Furthermore, training the BRB with accumulated clinical cases significantly improved diagnostic performance. Wu et al. [31] developed an Automatic Belief Rule Base (AutoBRB) model to accurately predict the complete pathological response in an interpretable way. The study compared five machine learning methods, including Support Vector Machine (SVM), Naive Bayes (NB), K-Nearest Neighbors (KNN), and Deep Neural Networks (DNN). The final experimental results showed that AutoBRB outperformed other available methods, demonstrating superior performance. Hossain et al. [32] designed a Belief Rule Base Expert System (BRBES) to diagnose tuberculosis with various types of uncertainties. The system's knowledge base was constructed by consulting experts and analyzing historical data from tuberculosis patients. Experiments involving data from 100 patients indicated that the BRBES results were more reliable than those of human experts and fuzzy rule-based expert systems. Han et al. [33] used a hierarchical BRB with a power set (H-BRBp) to diagnose lumbar spine diseases. Through multiple rounds of experimentation, H-BRBp maintained an accuracy rate of more than 90%, while machine learning algorithms such as Random Forest (RF), K-Nearest Neighbors (KNN), and Extreme Learning Machine (ELM) achieved accuracy rates between 70% and 80%. Due to its expert knowledge background, H-BRBp not only exhibits superior stability but also offers a degree of interpretability.

This paper explores a CKD diagnostic model based on traditional BRB, however, during the exploration process, there are three main drawbacks to the CKD diagnostic model based on traditional BRB. First, as CKD diagnostic technology evolves, renal function indicators may change, requiring experts to redesign the model to incorporate new indicators. Second, not all rules in the rule base of the CKD diagnostic model based on traditional BRB contribute positively; invalid and inefficient rules can reduce the model's clarity and interpretability and complicate optimization. Lastly, the interpretability of the CKD model based on traditional BRB is affected by the subjectivity of expert judgment during model creation, and random algorithmic adjustments can undermine the model's interpretability and trustworthiness of the results.

To address these issues, this paper proposes a new CKD diagnostic model based on DBRB-I, DBRB-I means an interpretable deep belief rule base. The CKD diagnostic model based on DBRB-I constructs a deep hierarchical structure layer by layer, effectively overcoming the scalability issues of the CKD diagnostic model based on traditional BRB. Guided by expert knowledge, the CKD diagnostic model based on DBRB-I simplifies the initial model through the rule reduction method aligned with expert judgment, thereby reducing the overall optimization difficulty. Furthermore, the CKD diagnostic model based on DBRB-I utilizes optimization algorithms that add new constraints to ensure transparent and reliable output results.

The main contributions of this paper include:

1) Integrating a deep belief rule base model into the diagnosis of CKD. "Deep" refers to the construction of a hierarchical structure, which allows the addition of new renal function indicators as needed. When a new indicator needs to be added, it can simply be incorporated by extending an existing layer, rather than redesigning the entire model from scratch as in the CKD diagnostic model based on traditional BRB.
2) Designing a new interpretable rule reduction method. Specifically, under the guidance of experts, efficient rules are extracted from the rule base in an interpretable manner. This approach can eliminate invalid and inefficient rules during the optimization process, thereby reducing the difficulty of optimizing the model.
3) Proposing a new optimization algorithm with constraints aligned with expert knowledge judgment. In practice, constraints are added to existing optimization algorithms under the guidance of expert knowledge to ensure that the algorithm's output consistently remains within the interpretable range set by experts.

The rest of this paper is organized as follows. Section II details the problems with the CKD diagnostic model based on BRB. Section III outlines the overall modeling process of the CKD diagnostic model based on DBRB-I, including the model construction, inference, optimization, and development processes. Section IV analyzes practical cases to validate the effectiveness of the proposed CKD diagnostic model based on DBRB-I. Section V summarizes the work and discusses its limitations.

II. PROBLEM DESCRIPTION

Before commencing this work, a fundamental prerequisite must be established: ensuring the trustworthiness of expert knowledge when constructing the CKD diagnostic model. Experts acquire their knowledge through extensive clinical experience and data analysis of CKD symptoms. Although expert knowledge has certain limitations in the accurate diagnosis of CKD, it must appropriately and reasonably guide the process of developing the CKD diagnostic model.

Problem 1: How to overcome the poor scalability issue of the CKD diagnostic model based on traditional BRB. In the practical CKD diagnostic environment, the renal function indicators entered into the model can change with advances in CKD diagnostic technology. Poor scalability can cause the model's diagnostic capabilities to lag current standards and fail to provide useful insights for experts. Therefore, a deep, layer-by-layer construction of the model can effectively address the scalability issues of the CKD diagnostic model based on traditional BRB.

$$\Gamma = \Psi\left(\chi_1, \chi_2, \ldots, \chi_n\right) \tag{1}$$

where Γ is the actual model structure of the CKD diagnostic model based on DBRB, χ_i $(i = 1, 2, \ldots, n)$ is the i-th renal function indicator entered into the model, n is the number of the renal function indicators entered into the model, and Ψ is the process of building the CKD diagnostic model based on DBRB.

Problem 2: How to reduce the optimization difficulty of the CKD diagnostic model based on DBRB. In the CKD diagnostic model based on DBRB, the initial parameters that need optimization are the rules, each of which is developed based on expert knowledge. The goal of optimizing these rules is to improve the accuracy of the model. However, not all rules in the rule base significantly impact the model's accuracy. In this paper, we define rules that have no impact on the model as invalid rules and rules that have minimal impact as inefficient rules. Although invalid and inefficient rules are also optimized as parameters in conventional models, they are essentially redundant in practice. Removing redundant parameters while keeping other conditions constant can reduce the overall difficulty of model optimization. Thus, simplifying rules based on expert knowledge can decrease the difficulty in optimizing the CKD diagnostic model based on DBRB.

$$\rho = \Phi\left(\rho_{\text{initial}}, Ef_k, E\right) \tag{2}$$

$$Ef_k = \Theta\left(Num_k, Per_k, \delta, E\right) \tag{3}$$

where ρ is the collection of parameters that need to be involved in the model optimization process after simplifying the rules based on expert knowledge, Φ is the process of simplifying the rules, ρ_{initial} is the collection of initial parameters that need to be involved in the model optimization process, Ef_k is the corresponding efficiency factor of the k-th rule, E is the expert knowledge involved in the construction of the model, Θ is the process of calculating the efficiency factor, Num_k is the number of times that the k-th rule has been activated, Per_k is the percentage of the smaller of all the activation weights in the k-th rule, and δ is the collection of parameters required to calculate the efficiency factor.

Problem 3: How to maintain the interpretability of the CKD diagnostic model based on DBRB. Using expert knowledge in parameter setting and a transparent reasoning process yields the initial CKD diagnostic model based on DBRB, offering interpretability that experts can understand and trust. However, optimization processes often prioritize high accuracy, potentially compromising interpretability. Consequently, the model output may contradict expert knowledge, rendering it unsuitable for practical CKD diagnosis. Therefore, developing an optimization process that aligns with expert judgment can preserve the interpretability of the CKD diagnostic model based on DBRB.

$$\rho_{\text{best}} = \Xi\left(\rho, \xi, E\right) \tag{4}$$

$$\text{Interpretability Constraints: } \{\xi \mid \xi_1, \xi_2, \ldots, \xi_8\} \tag{5}$$

where ρ_{best} is the collection of the best parameters after the model optimization process, ξ is the collection of interpretability constraints of the model, and Ξ is the model optimization process.

III. THE CKD DIAGNOSTIC MODEL BASED ON DBRB-I

This section describes the overall modeling process of the CKD diagnostic model based on DBRB-I and effectively addresses the three key issues raised in Section II. Section III-A details the construction of the CKD diagnostic model based on DBRB-I, including the basic structure of the model in Section III-A1, selection of renal function indicators in Section III-A2, rules in Section III-A3, and the rule reduction method based on the efficiency factor in Section III-A4. Section III-B explains how to derive the final results of the CKD diagnostic model based on DBRB-I using the inference method. Section III-C provides a detailed presentation of the model's constraints and optimization process. Section III-D describes the relationships and workflows between the various parts of the CKD diagnostic model based on DBRB-I.

A. BUILDING THE CKD DIAGNOSTIC MODEL BASED ON DBRB-I

1) The basic structure of the CKD diagnostic model based on DBRB-I

In the CKD diagnostic environment, renal function indicators are chosen based on expert knowledge or data mining techniques. These chosen indicators are then input into the CKD diagnostic model based on traditional BRB. Through continuous optimization, the model is refined to ensure its final output effectively diagnoses whether a patient has CKD. The basic structure of the CKD diagnostic model based on traditional BRB is shown in Figure 1.

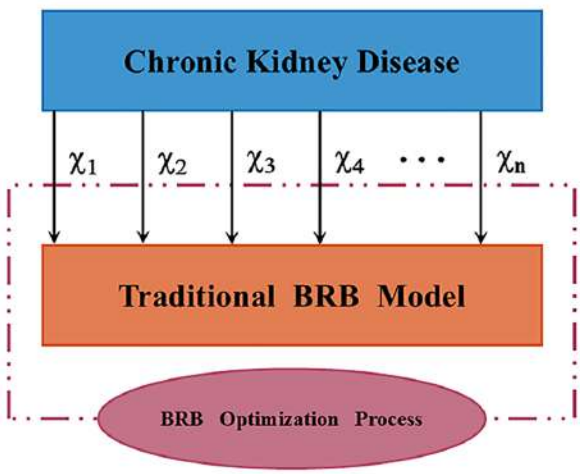

FIGURE 1. Structure of the CKD diagnostic model based on traditional BRB.

As new data and knowledge accumulate, CKD diagnostic models need regular updates and maintenance to maintain their effectiveness and accuracy in new situations. Furthermore, in the CKD diagnostic model based on traditional BRB, the number of rules depends on the number of renal function indicators and their reference values. For instance, with 5 reference values per indicator, a model with 2 indicators has

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25 rules. However, if the number of indicators increases to 3 while keeping the reference values constant, the number of rules jumps to 125. Since CKD is influenced by multiple renal function indicators, an increase in indicators leads to an exponential growth in the number of rules, causing a rule explosion problem. These factors highlight the limited scalability of the CKD diagnostic model based on traditional BRB.

The CKD diagnostic model based on DBRB-I extends from the construction principles of the CKD diagnostic model based on traditional BRB. It involves selecting renal function indicators first, then arranging them in a sequence based on their impact, and finally building the model layer by layer with interpretability constraints during the optimization process. This approach effectively addresses the scalability issues of the CKD diagnostic model based on traditional BRB while maintaining its interpretability. The basic structure of the CKD diagnostic model based on DBRB-I is shown in Figure 2.

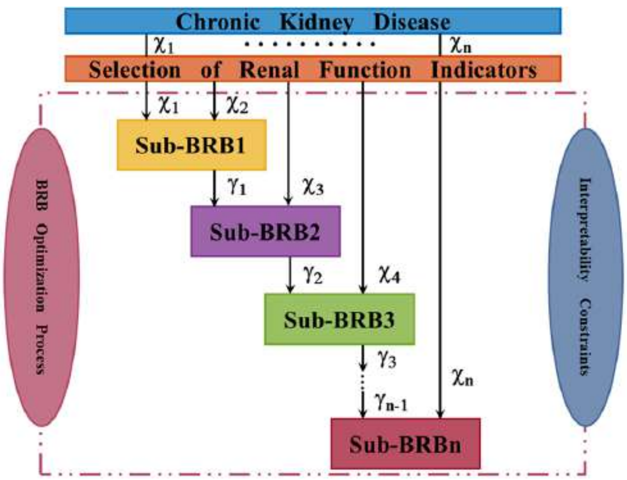

FIGURE 2. Structure of the CKD diagnostic model based on DBRB-I.

By comparing Figure 1 with Figure 2, it is evident that there is a significant difference in the complexity of the model between the CKD diagnostic model based on traditional BRB and DBRB-I. The complexity of BRB-related models depends mainly on the number of rules, which can be calculated using (6). Assuming that the CKD diagnostic model based on BRB requires four renal function indicators, with five reference values for each indicator, the calculation shows that the CKD diagnostic model based on traditional BRB has 625 rules, while the CKD diagnostic model based on DBRB-I contains only 75 rules. This demonstrates that the CKD diagnostic model based on DBRB-I significantly reduces the complexity of the model compared to the model based on traditional BRB.

$$l = Lay \times Val^{ln} \tag{6}$$

where l is the total number of rules in the BRB model, Lay is the number of layers of the BRB model, Val is the number of reference values, and ln is the number of renal function indicators entered in each layer.

2) Selection of renal function indicators for the CKD diagnostic model based on DBRB-I

The selection of renal function indicators in the CKD diagnostic model based on DBRB-I involves two key aspects. First, it is essential to scientifically determine the minimal number of renal function indicators to include in the model, under expert guidance, ensuring that the final performance of the model remains unaffected. This approach not only reduces the number of input parameters but also simplifies the overall complexity of the model, enhancing its interpretability and making it more user-friendly for experts in practical diagnoses. Second, since the CKD diagnostic model based on DBRB-I is constructed layer by layer, with each layer's input depending on the output of the previous layer, it is critical to arrange the renal function indicators in a sequence that accurately reflects their influence on the model.

When experts cannot determine the minimal number and sequence of renal function indicators, a gray-box model that explains the impact of these indicators, such as Random Forest, can be used. Random Forest consists of multiple decision trees, allowing experts to gain an intuitive understanding of the overall decision-making process by examining the decision paths of each tree [34]. It assesses the contribution of each renal function indicator to the diagnosis of CKD through its importance, helping experts identify which indicators are the most critical for the model diagnosis and further understand the model's decision-making process [35].

3) Rules of the CKD diagnostic model based on DBRB-I

BRB models are conducted through IF-THEN rules, which offer an easily understandable process and traceable results. The core of BRB is to use IF-THEN rules to represent the relationship between input information and corresponding output results, reflecting expert knowledge. In IF-THEN rules, IF represents the cause and THEN indicates the consequence. Simply put, if a condition is met, a corresponding outcome will occur. The foundational rules for constructing the CKD diagnostic model based on DBRB-I are also IF-THEN rules, and their representation is as follows:

$$\begin{aligned} &R_k : \text{If } \chi_1 \text{ is } A_1 \wedge \chi_2 \text{ is } A_2 \wedge \ldots \wedge \chi_n \text{ is } A_n, \\ &\text{Then } \gamma \text{ is } \{(G_1, \beta_1), (G_2, \beta_2), \ldots, (G_m, \beta_m)\} \\ &\text{With } \varpi_k \text{ and } \alpha_1, \alpha_2, \ldots, \alpha_n \end{aligned} \tag{7}$$

where R_k $(k = 1, 2, \ldots, l)$ is the k-th rule, A_i is the reference value corresponding to the i-th renal function indicator in the k-th rule, γ is the output result of the model, G_j $(j = 1, 2, \ldots, m)$ is the j-th evaluation grade of the output result in the k-th rule, m is the number of the output results, β_j is the belief corresponding to the j-th evaluation grade of the output result in the k-th rule, ϖ_k is the rule weight corresponding to the k-th rule, and α_i is the attribute weight corresponding to the i-th renal function indicator in the k-th rule.

4) The rule reduction method of the CKD diagnostic model based on DBRB-I

Current BRB rule reduction methods mostly categorize rules simply as valid and invalid ones, without further distinguishing valid rules into inefficient and efficient categories. Inefficient rules have minimal impact on the model's final decision but consume unnecessary resources (memory and time). Retaining only efficient rules in the BRB model significantly improves readability, allowing experts to better understand how the model makes decisions. Some rule reduction methods use mathematical models to analyze each rule's impact on output, clarifying rule types. For example, Han et al. [36] use local sensitivity analysis (LSA) to determine the efficiency of the rules, while Yang et al. [37] use data envelope analysis (DEA) for the evaluation of efficiency. However, these methods do not fully integrate expert knowledge, only comparing results with expert judgments in case studies.

In the rule reduction method proposed in this paper, expert knowledge is incorporated in two ways. First, it is used to establish the conditions for rule reduction. Second, it guides the calculation of the efficiency factor for each rule. This method, based on the core IF-THEN rules, determines whether a rule is invalid, inefficient, or efficient by assessing which type of condition the efficiency factor satisfies. The representation of the rule reduction method is as follows:

$$\begin{aligned}&\text{If } Ef_k \text{ is } C_{\text{invalid}} \text{ or } C_{\text{inefficient}} \text{ or } C_{\text{efficient}},\\&\text{Then } R_k \text{ is } R_{\text{invalid}} \text{ or } R_{\text{inefficient}} \text{ or } R_{\text{efficient}}\end{aligned} \tag{8}$$

where C_{invalid} is the condition specified by the expert to judge the invalid rule, $C_{\text{inefficient}}$ is the condition specified by the expert to judge the inefficient rule, $C_{\text{efficient}}$ is the condition specified by the expert to judge the efficient rule, R_{invalid} is the invalid rule in the model, $R_{\text{inefficient}}$ is the inefficient rule in the model, and $R_{\text{efficient}}$ is the efficient rule in the model.

Here, Ef_k denotes the contribution of the k-th rule to the overall performance of the model. Measurement of Ef_k involves two aspects: first, the number of times the k-th rule is used or activated in the CKD diagnostic environment; and second, the degree to which the k-th rule is activated in the CKD diagnostic environment. The specific calculation process of the efficiency factor is described as follows:

Step 1: Counting the activation frequency of the k-th rule Num_k.

Step 2: Evaluating the magnitude of the activation weights for the k-th rule.

$$\text{If } Aw_k^t < \delta_1 \text{ and } Aw_k^t \neq 0, \quad \text{Then } Aw_k^t \text{ is } Aw_k^{low}. \tag{9}$$

where Aw_k^t $(t = 1, 2, \ldots, s)$ is the t-th activation weight in the k-th rule, s is the total number of activation weights that are not 0 in the k-th rule, δ_1 is the threshold specified by the expert to determine the importance of the activation weight, and Aw_k^{low} is the activation weight that is judged to be the smallest in the k-th rule.

Step 3: Counting the number of smaller activation weights for the k-th rule Num_k^{low}.

Step 4: Calculating the proportion of smaller activation weights within the k-th rule.

$$Per_k = \frac{Num_k^{low}}{Num_k} \tag{10}$$

Step 5: Computing the efficiency factor.

$$Ef_k = \begin{cases} Per_k + Num_k \times \delta_3, & Num_k \leq \delta_2 \\ Per_k + Num_k \times \delta_4, & Num_k > \delta_2 \end{cases} \tag{11}$$

where δ_2 is a threshold value, and δ_3 and δ_4 are values specified by the expert to calculate the efficiency factor.

Following this process, the method integrates expert knowledge into both the establishment of conditions and the calculation of the efficiency factor, ensuring a comprehensive approach to rule reduction in the CKD diagnostic model based on DBRB-I.

B. INFERENCE OF THE CKD DIAGNOSTIC MODEL BASED ON DBRB-I

Yang et al. [19] proposed a unique evidence-reasoning rule based on reliability-weighted belief distributions. The ER rule establishes a generic conjunctive probability reasoning method, which is the inference method of BRB. The CKD diagnostic model based on DBRB-I also uses this transparent and easy-to-understand ER inference method. The specific process of the ER inference method is described as follows:

Step 1: Transform different types of input information into belief distributions.

$$S(x_i) = \{(A_{i,e}, a_{i,e})\}, \quad e = 1, 2, \ldots, f\} \tag{12}$$

$$a_{i,e} = \frac{A_{i,e+1} - x_i}{A_{i,e+1} - A_{i,e}}, a_{i,e+1} = 1 - a_{i,e}, A_{i,e} \leq x_i \leq A_{i,e+1} \tag{13}$$

$$a_{i,b} = 0, \quad b = 1, 2, \ldots, f, \quad b \neq e, e+1 \tag{14}$$

where $A_{i,e}$ is the e-th reference value of the i-th renal function indicator and $a_{i,e}$ is the degree of matching of input information x_i with respect to $A_{i,e}$.

Step 2: Calculate the activation weight of the input information for the k-th belief rule.

$$Aw_k = \frac{\mu_k \prod_{i=1}^{n_k} \left(a_{i,e}^k\right)^{\bar{\theta}_i}}{\sum_{c=1}^{l} \mu_c \prod_{i=1}^{n_c} \left(a_{i,e}^c\right)^{\bar{\theta}_i}} \tag{15}$$

$$\bar{\theta}_i = \frac{\theta_i}{\max\{\theta_i\}} \tag{16}$$

where $\bar{\theta}_i$ is the normalized weight of the i-th renal function indicator.

Step 3: Generate the corresponding belief degree of the output result by the ER analytical algorithm.

$$\begin{aligned}\beta_j = &\frac{\prod_{k=1}^{l} \left(Aw_c \beta_{j,c} + 1 - Aw_c \sum_{i=1}^{m} \beta_{i,c}\right)}{\lambda - \prod_{c=1}^{l} (1 - Aw_c)} \\ &- \frac{\prod_{c=1}^{l} \left(1 - Aw_c \sum_{i=1}^{m} \beta_{i,c}\right)}{\lambda - \prod_{c=1}^{l} (1 - Aw_c)}\end{aligned} \tag{17}$$

$$\lambda = \sum_{j=1}^{m} \prod_{c=1}^{l} \left(Aw_c \beta_{j,c} + 1 - Aw_c \sum_{i=1}^{m} \beta_{i,c} \right) - (m-1) \prod_{c=1}^{l} \left(1 - Aw_c \sum_{i=1}^{m} \beta_{i,c} \right) \tag{18}$$

Step 4: Combine all the rules to obtain the final result of the CKD diagnostic model based on DBRB-I.

$$S\left(A'\right) = \left\{ \left(G_j, \beta_j\right), j = 1, 2, \ldots, m \right\} \tag{19}$$

$$u\left(S\left(A'\right)\right) = \sum_{j=1}^{m} u\left(G_j\right) \beta_j \tag{20}$$

where A' is the actual input to the CKD diagnostic model based on DBRB-I, $u\left(G_j\right)$ is the utility of G_j and $u\left(S\left(A'\right)\right)$ is the ultimate expected utility of the CKD diagnostic model based on DBRB-I.

C. OPTIMIZATION OF THE CKD DIAGNOSTIC MODEL BASED ON DBRB-I

The integration of expert knowledge enhances the overall interpretability of the CKD diagnostic model based on DBRB-I. However, interpretability and accuracy often cannot be simultaneously optimized. This is because interpretability reflects the limitations of expert knowledge, and an overemphasis on it may diminish the model's problem-solving capabilities. To strike a balance between interpretability and accuracy, an optimization process incorporating interpretability constraints is essential. The CKD diagnostic model based on DBRB-I, with these constraints, can foster greater trust and understanding among both experts and patients, making it more suitable for real-world diagnostic applications. Therefore, based on the general interpretability guidelines established by Cao et al. [38], this paper proposes interpretability standards for the CKD diagnostic models based on DBRB-I. The specific definitions and descriptions of these interpretability standards are as follows:

Constraint 1: Each renal function indicator used in the CKD diagnostic model based on DBRB-I must have a clear semantic value.

Renal function indicators have their representation methods in real life, including quantitative methods using numerical values and qualitative methods that are difficult to express numerically. Although renal function indicators are quantitative or qualitative, they should be represented in the CKD diagnostic model based on DBRB-I. The model should set reference value ranges for each renal function indicator, with each indicator having at least two reference values. In addition, different renal function indicators have different medical meanings, which means that entering different indicators will affect the final diagnostic result. Therefore, different renal function indicators should be differentiated in the CKD diagnostic model, each having its reference value range. Finally, it is essential to ensure that each patient sample input into the CKD diagnostic model is associated with at least one reference value. The specific reference value ranges can be described as follows:

$$\begin{aligned} &\forall 1 \le \sigma \le f, \quad f \ge 2, \quad \exists x_0 \in P, \quad a_{i,\sigma}(x_0) = 1 \\ &\forall 1 \le e \le f, \quad f \ge 2, \quad x \in P, \quad 0 \le a_{i,e}(x) \le 1 \end{aligned} \tag{21}$$

where f is the total number of reference values and x_0 is a fixed value in the feasible domain P.

Constraint 2: The structure and rules of the CKD diagnostic model based on DBRB-I must have practical significance.

The CKD diagnostic model based on DBRB-I uses a hierarchical structure. To ensure expert trust and practical application in the real CKD diagnostic environment, the new layer with the additional renal function indicator should improve diagnostic performance compared to the previous layer. The rules are central to the CKD diagnostic model based on DBRB-I. If the rules lack practical significance, the entire model becomes meaningless. Rules are represented in the model through various parameters, including belief, rule weight, attribute weight, activation weight, and the efficiency factor. The specific ranges of these parameters are described as follows:

$$\{\beta, \varpi, \alpha, Aw\} \in [0, 1], \quad Ef \ge 0 \tag{22}$$

Constraint 3: The CKD diagnostic model based on DBRB-I must have a complete and as simple as possible rule base.

The rule base of the CKD diagnostic model based on DBRB-I must be comprehensive to ensure that every possible input value has at least one matching reference, which means that at least one rule can be activated and used. This ensures that the model can diagnose any patient with CKD in the real environment. In addition, a simpler rule base makes it easier for experts to understand and use the model. The rule base can be described as follows:

$$\forall x \in P, \quad \exists 1 \le \sigma \le l, \quad Aw_\sigma(x) > 0 \tag{23}$$

where x is any sample of patients that may be entered into the CKD diagnostic model based on DBRB-I.

Constraint 4: The rules in the CKD diagnostic model based on DBRB-I must be consistent with each other.

The rule base of the CKD diagnostic model based on DBRB-I consists of several rules that must remain consistent to avoid contradictions. Otherwise, it would be difficult to determine which rule influenced the final diagnostic result, thereby reducing the model's interpretability. By having experts develop the rules, consistency can be ensured, thus enhancing both the model's interpretability and the reliability of the diagnostic results. The relationship between the rules can be described as follows:

$$R_1 \cap R_2 \cap \cdots \cap R_l = 0 \tag{24}$$

Constraint 5: The information transformation in the CKD diagnostic model based on DBRB-I must be reasonable.

The CKD diagnostic model based on DBRB-I must ensure the completeness of the initial information and the equivalence of the transformation process. This is because the first step of the inference process is to convert different types of input information into belief distributions. The correctness and rationality of this step determine the effectiveness of the entire inference process, influencing whether the final diagnostic result can gain the trust of experts. To ensure the rationality of the information transformation, the CKD diagnostic model based on DBRB-I employs an ER method based on rules and utilities, which effectively ensures equivalence within the belief structure and facilitates a reasonable information transformation. The information transformation process in the ER method can be described by (12) - (14).

Constraint 6: The inference process of the CKD diagnostic model based on DBRB-I should be transparent and clear.

A transparent and clear inference process means that both experts and patients can understand each step of the inference in the CKD diagnostic model based on DBRB-I, thereby enhancing trust in the diagnostic results. The ER inference method used in the CKD diagnostic model based on DBRB-I provides this transparency, effectively ensuring the model's interpretability. The specific inference steps are described by (15) - (20).

Constraint 7: The CKD diagnostic model based on DBRB-I must make full use of expert knowledge.

Expert knowledge forms the foundation of interpretability in the CKD diagnostic model based on DBRB-I, playing a crucial role throughout its development. In the initial phase, expert knowledge is employed to set the initial parameters and generate the first set of rules, establishing a framework for subsequent reasoning. During the rule reduction process, experts assess and select the most efficient rules, enhancing both the model's performance and interpretability. Additionally, expert knowledge is incorporated into the optimization algorithm, with constraints applied to ensure that expert judgment is fully leveraged, thereby maximizing the model's interpretability. Expert knowledge is involved throughout the CKD diagnostic model based on DBRB-I in the building process Ψ, the rule reduction Φ, the calculation of the efficiency factor Θ, and the optimization process Ξ, and they are described as follows:

$$E \xrightarrow{\text{involved}} \{\Psi, \Phi, \Theta, \Xi\} \tag{25}$$

Constraint 8: The optimized parameters of the CKD diagnostic model based on DBRB-I should not contradict the judgments of expert knowledge.

Unlike other CKD diagnostic models based on black-box methods, the CKD diagnostic model based on DBRB-I uses parameters that represent meaningful rules, making it easier for CKD patients to accept the model's diagnostic decisions. However, the optimization algorithm itself is stochastic, which can lead to optimized parameters that contradict expert judgment. To address this issue, the CKD diagnostic model based on DBRB-I incorporates constraint operation into the optimization algorithm to ensure that the optimized parameters always fall within the range allowed by expert knowledge. The specific range can be defined on the basis of the expert's experience and domain knowledge, ensuring that the model's outputs are both theoretically and practically consistent and interpretable. The specific range can be described as follows:

$$E_{\text{lower}} \leq \rho_{\text{best}} \leq E_{\text{upper}} \tag{26}$$

where E_{upper} and E_{lower} are the upper and lower bounds of the optimization parameters set by the expert.

In simple terms, the optimization process involves using optimization algorithms to continually refine the model parameters to achieve the best possible solution, thereby maximizing model performance. In the application of BRB-related models, common optimization algorithms include Particle Swarm Optimization (PSO), Sequential Quadratic Programming (SQP), Differential Evolution (DE), and Projected Covariance Matrix Adaptation Evolution Strategy (P-CMA-ES) [39]. Compared to other algorithms, the main advantage of P-CMA-ES lies in its combination of the covariance matrix adaptation mechanism and projection operations, which enhances its ability to handle complex high-dimensional optimization problems [40]. Improve global search efficiency, reduce dimensional redundancy, and avoid local optima by adaptively adjusting the covariance matrix, ultimately improving optimization performance and convergence speed [41]. As a result, P-CMA-ES is well suited for more complex optimization tasks in practical applications and has been widely used in various BRB studies [42]–[44]. In addition, several studies have compared these optimization algorithms. For example, Zhou et al. [45] compared PSO, SQP, and P-CMA-ES in the study of the hidden power set BRB, finding that P-CMA-ES provided higher accuracy. Liu et al. [46] compared PSO, DE, and P-CMA-ES in the study of BRB with automated construction, showing that P-CMA-ES had better robustness. Cao et al. [38] compared DE and P-CMA-ES in the study of interpretable BRB, finding that P-CMA-ES offered stronger interpretability. Based on these findings, this paper selects P-CMA-ES as the optimization algorithm for the CKD diagnostic model based on DBRB-I.

However, the P-CMA-ES algorithm is an iterative process, in which each sampling operation is closely related to the previous results. After hundreds of iterations, the final model parameters might contradict expert knowledge, potentially compromising the model interpretability [47]. Therefore, to maintain the interpretability of the CKD diagnostic model based on DBRB-I, constraint operation has been added to the original P-CMA-ES. The modified P-CMA-ES is illustrated in Figure 3. The specific steps of the P-CMA-ES algorithm in the CKD diagnostic model based on DBRB-I are described as follows:

Step 1 (Sampling operation): Generate the search population according to Constraint 7.

$$Q^0 = \{\beta_{1,1}, \ldots, \beta_{n,l}, \varpi_1, \ldots, \varpi_l, \alpha_1, \ldots, \alpha_n\} \tag{27}$$

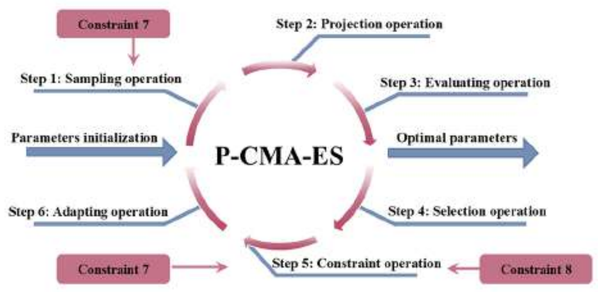

FIGURE 3. Optimization process of the modified P-CMA-ES.

$$Q_h^{g+1} = Z^g + \epsilon^g D(0, M^g), \quad h = 1, 2, \ldots, w \tag{28}$$

$$Z^g = \begin{cases} E, & \text{if } g = 1 \\ Z^g, & \text{if } g \neq 1 \end{cases} \tag{29}$$

where Q^0 is the parameter set by the expert that needs to be optimized, Q_h^{g+1} is the h-th individual of the search population updated to the $g+1$ generation, Z^g is the mean of individuals in the current search population and the central reference point for the new search population, ϵ^g is the step size of the g-th generation, D is the normal distribution, which is the probability distribution used to generate new individuals, M^g is the covariance matrix of the g-th generation, and w is the number of total generations.

Step 2 (Projection operation): If the candidate individuals from the Sampling operations do not satisfy the equality constraints, project these candidates onto the feasible region hyperplane to find an individual that meets all constraints.

$$\begin{aligned} &Q_h^{g+1}(1 + v \times (u-1) : v \times u) \\ &= Q_h^{g+1}(1 + v \times (u-1) : v \times u) - \frac{T^n}{T \times T^n} \\ &\times Q_h^{g+1}(1 + v \times (u-1) : v \times u) \times T \end{aligned} \tag{30}$$

$$TQ_h^g(1 + v \times (u-1) : v \times u) = 1, \quad u = 1, 2, \ldots, D+1 \tag{31}$$

where v is the number of variables of the equality constraints in the individual Q_h^{g+1}, u is the number of equality constraints, T^n is the parameter vector, and (31) is the hyperplane expression.

Step 3 (Evaluating operation): To improve model diagnostic accuracy, the parameters of the CKD diagnostic model based on DBRB-I are optimized using training data. An objective optimization function is set up to assess the difference between the diagnostic results of the CKD model based on DBRB-I and the actual CKD diagnostic results. The objective optimization function for the CKD diagnostic model based on DBRB-I is described as follows:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} (y_{\text{true}}^i - y_{\text{predict}}^i)^2 \tag{32}$$

where N is the number of training samples, y_{true}^i is the true value of the CKD diagnosis, and $y_{predict}^i$ is the predicted value of the CKD model.

Step 4 (Selection operation): Calculate the values of the objective optimization function for the candidate individuals and sort them, then select the optimal solutions from the entire population.

$$\text{MSE}\left(Q_{1:w}^{g+1}\right) \leq \cdots \leq \text{MSE}\left(Q_{h:w}^{g+1}\right) \leq \cdots \leq \text{MSE}\left(Q_{w:w}^{g+1}\right) \tag{33}$$

$$Z^{g+1} = \sum_{h=1}^{\eta} \omega_h Q_{h:w}^{g+1} \tag{34}$$

where η is the population size of the progeny and ω_h is the weighting factor.

Step 5 (Constraint operation): Apply Constraint 7 and Constraint 8 to the center point of the generated search population to ensure that the next generation remains within the range permitted by expert knowledge.

$$\text{If } d(Z^1, Z^{g+1}) > \tau, \text{ Then } Z^{g+1} = Z^1 \times p + Z^{g+1} \times q \tag{35}$$

where d is the distance between Z^1 with Z^{g+1}, τ is the maximum distance set by the expert, p is the expert reliability set by the expert and q is the reliability of the optimization algorithm set by the expert.

Step 6 (Adapting operation): Update the covariance matrix based on the selected optimal solutions. The updated covariance matrix will be used as the basis for generating the search population in the next Sampling operation.

$$\begin{aligned} M^{g+1} &= (1 - r_1 - r_2)M^g + r_1 P_r^{g+1}(P_r^{g+1})^n \\ &+ r_2 \sum_{h=1}^{\delta} \omega_h \left(\frac{K_{h:w}^{g+1} - \phi^g}{\epsilon^g}\right)\left(\frac{K_{h:w}^{g+1} - \phi^g}{\epsilon^g}\right)^n \end{aligned} \tag{36}$$

where r_1 and r_2 is the learning rate, P_r^{g+1} is the evolution path of the $g+1$ generation, $K_{h:w}^{g+1}$ is the h-th parameter vector of vector w in the $g+1$ generation and ϕ^g is the number of descendants in the g generation.

Overall, based on Constraints 7 and 8, as well as the specific steps in the optimization process, expert knowledge primarily influences the first and fifth steps of optimization. In the first step, the expert defines the central reference point of the initial search population. Unlike a random central reference point, the expert's involvement enhances the interpretability of the optimization process. In the fifth step, the expert sets the parameters necessary for the constraint operation, effectively constraining the central reference point of the new search population. This ensures that each optimization starts within the interpretable range defined by the expert. Through these two steps, the optimization process effectively preserves expert judgment, safeguarding the model's interpretability. This not only ensures that the model's decisions align with expert knowledge but also enhances its transparency and credibility.

D. DEVELOPMENT OF THE CKD DIAGNOSTIC MODEL BASED ON DBRB-I

The CKD diagnostic model based on DBRB-I consists of five main components: Random Forest, DBRB-I, rule reduction, P-CMA-ES, and expert knowledge. These components work together to complete the CKD diagnosis task. Their relationships and workflow are shown in Figure 4. The implementation steps of the CKD diagnostic model based on DBRB-I and the role of expert knowledge are as follows:

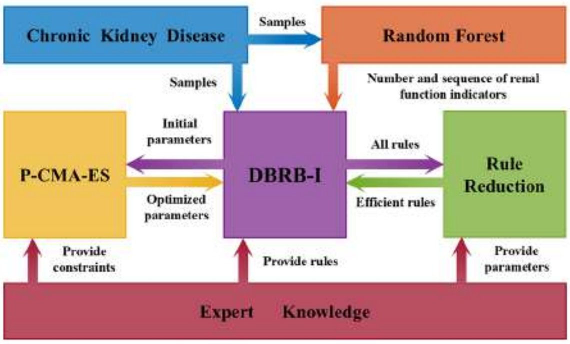

FIGURE 4. Flowchart of the CKD diagnostic model based on DBRB-I.

First, the Random Forest method is used to select the minimal number of renal function indicators and their input order from the CKD samples, as detailed in Section III-A2. Next, based on CKD samples and information from renal function indicators, the initial CKD diagnostic model based on DBRB-I is constructed. The specific structure can be shown in Section III-A1. The rules for this model are provided by expert knowledge, supporting the model inference and decision-making process, as outlined in Section III-A3. Then, the efficiency factors of all rules in the model are calculated, with expert knowledge providing the parameters for this calculation. A rule reduction method is used to select efficient rules, which are then integrated into the model, as described in Section III-A4. Finally, the P-CMA-ES algorithm is employed to optimize the model's parameters, with experts providing constraints to guide the search range for the parameters, ensuring that the optimization process enhances both the model's accuracy and interpretability, as detailed in Section III-C. It is important to note that, since the CKD diagnostic model based on DBRB-I is constructed incrementally, both optimization and rule reduction must be carried out layer by layer.

Expert knowledge serves as the cornerstone of the CKD diagnostic model based on DBRB-I and plays a crucial role in several key steps. First, expert knowledge helps to build the initial rule base, ensuring that the model can effectively diagnose CKD based on domain expertise. Then, after generating all possible rules, expert knowledge intervenes in the rule reduction process to simplify the rule base, eliminating invalid and inefficient rules, and thus reducing the optimization difficulty while maintaining interpretability. Finally, during the model optimization phase, expert knowledge provides constraints to guide the P-CMA-ES optimization, ensuring that the optimization process enhances both the accuracy and interpretability of the model.

IV. CASE STUDY

The dataset used to validate the performance of the CKD diagnostic model based on DBRB-I comes from the KAGGLE at https://www.kaggle.com/datasets/abhia1999/chronic-kidney-disease. The CKD dataset aims to predict whether patients have Chronic Kidney Disease (CKD) based on the renal function indicators provided. It includes samples from 400 patients, each with complete information on 13 renal function indicators and one actual diagnostic result. Among these 400 patients, 250 have CKD while 150 do not. To this day, many researchers continue to explore CKD diagnostic methods and models using this dataset.

A. ANALYSIS AND SELECTION OF RENAL FUNCTION INDICATORS

In the experiment, the Random Forest method was used to analyze CKD data samples from 400 patients, selecting the minimal number and priority order of renal function indicators to be entered into the CKD diagnostic model based on DBRB-I. This approach ensures that the final set of selected renal function indicators can maintain diagnostic accuracy while minimizing the complexity of the model and improving efficiency. The experiment used 80% (i.e., 320 patients) of the data as the training set and 20% (i.e., 80 patients) as the test set. The results show that when the four most important renal function indicators are selected, the model performs optimally while maintaining the lowest complexity. Table 1 shows a comparison of the experimental results for different numbers of renal function indicators selected by the Random Forest method.

TABLE 1. Comparison of performance metrics for different numbers of renal function indicators.

Number of renal function indicators	Accuracy	Precision	Recall	F1 score
3	0.8663	0.8622	0.8021	0.8302
4	**0.9000**	**0.8929**	**0.8333**	**0.8621**
5	0.8775	0.8719	0.8125	0.8405
7	0.8888	0.8821	0.8229	0.8518
10	0.8775	0.8706	0.8125	0.8405
13	**0.9000**	**0.8929**	**0.8333**	**0.8621**

Figure 5 shows the feature importance scores for renal function indicators in the Random Forest method. Hemoglobin has a feature importance score of 0.2762, contributing 28% to the model; Specific Gravity has a score of 0.1632, contributing 16%; Albumin has a score of 0.1452, contributing 15%; Serum Creatinine has a score of 0.1421, contributing 14%. The remaining 9 combined renal function

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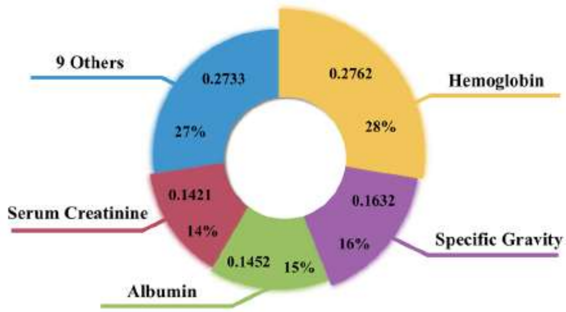

FIGURE 5. The feature importance scores for all renal function indicators.

indicators have a total feature importance score of 0.2733, which contributes only 27% to the model.

The final renal function indicators selected by the random forest for entry into the CKD diagnostic model based on DBRB-I are Hemoglobin, Specific Gravity, Albumin and Serum Creatinine. Figure 6 shows the CKD diagnostic model based on DBRB-I, which has a three-layer structure and is constructed according to the four indicators and sequences mentioned above.

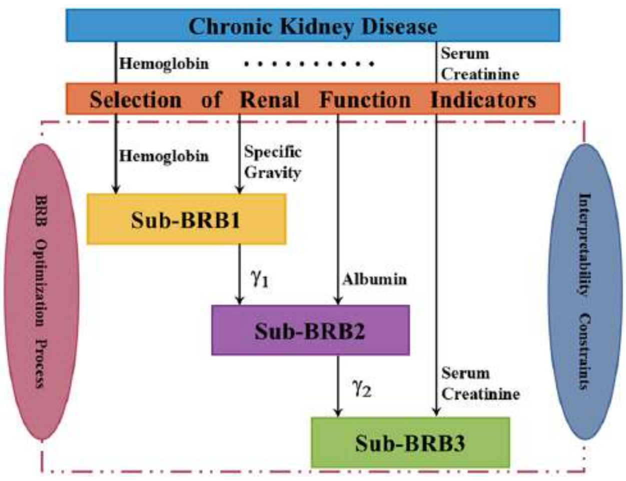

FIGURE 6. Actual structure of the CKD diagnostic model based on DBRB-I.

B. MODEL BUILDING

The reference values for the renal function indicators used in the CKD diagnostic model based on DBRB-I are defined by experts, with each indicator classified into five levels: VL (Very Low), L (Low), M (Moderate), H (High), and VH (Very High). Table 2 displays the meanings and specific values for each of these five levels.

Table 3 shows the diagnostic results obtained by the CKD diagnostic model based on DBRB-I, where "Health" is indicated by 0 and "CKD" is indicated by 1.

Appendix A contains the initial parameters set by expert knowledge for each sub-layer of the CKD diagnostic model based on DBRB-I, including the initial rule weights and the initial belief distributions. In all appendixes, "RW" is the rule weight, "RFI" is the renal function indicator, and the two lines are used to differentiate between different sub-BRBs.

TABLE 2. Reference values of renal function indicators.

Renal function indicator	VL	L	M	H	VH
Hemoglobin	3.1	10.9	13	14.8	18.7
Specific Gravity	1.005	1.011	1.016	1.021	1.026
Albumin	0	0.5	1.5	2.5	5.5
Serum Creatinine	0.4	1	1.5	3.1	76.1

TABLE 3. Reference values of diagnostic results.

diagnostic result	Health	CKD
Model output value	0	1

C. MODEL PERFORMANCE EVALUATION

1) Optimized difficulty assessment

Experts have set the criteria for the types of rules in the CKD diagnostic model based on DBRB-I, as described below:

$$\begin{aligned} C_{\text{invalid}}:&\quad Ef_k = 0;\\ C_{\text{inefficient}}:&\quad 0 < Ef_k < 1;\\ C_{\text{efficient}}:&\quad Ef_k \geq 1 \end{aligned} \tag{37}$$

Based on the above conditions, each type of rule in the CKD diagnostic model based on DBRB-I can be determined. Figure 7 shows the efficiency factor for each rule in Sub-BRB1, Sub-BRB2, and Sub-BRB3 of the CKD diagnostic model based on DBRB-I. The red line in the figure is crucial to distinguishing between inefficient and efficient rules. Appendix B lists the efficiency factor of the three sub-layers calculated in Section III-A4.

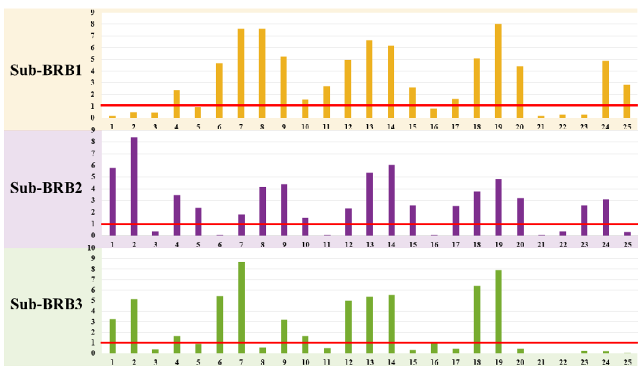

FIGURE 7. The efficiency factor of each rule in three sub-layers.

Table 4 shows the initial number of rules in the CKD diagnostic model based on DBRB-I, the number of efficient rules after rule reduction, and the efficient rule that must undergo the optimization process. Figure 8 compares the number of rules in each sub-layer of the CKD diagnostic model based on DBRB-I.

TABLE 4. The number of rules in the CKD diagnostic model based on DBRB-I

Sub-layer	Number of initial rules	Number of efficient rules	Efficient rule
Sub-BRB1	25	17	Rule 4,6,7,8,9,10,11,12,13, 14,15,17,18,19,20,24,25
Sub-BRB2	25	18	Rule 1,2,4,5,7,8,9,10,12,13, 14,15,17,18,19,20,23,24
Sub-BRB3	25	12	Rule 1,2,4,6,7,9, 10,12,13,14,18,19

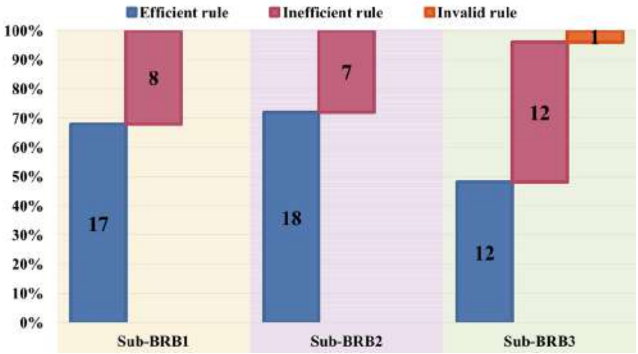

FIGURE 8. Comparison of the number of different types of rules in three sub-layers.

By analyzing the activation weights of the rules in each sub-layer of the CKD diagnostic model based on DBRB-I, the effectiveness and reliability of using the efficiency factor to determine rule types can be further demonstrated. Figures 9, 10, and 11 show the activation weights of each rule in Sub-BRB1, Sub-BRB2, and Sub-BRB3, respectively. Red points indicate that the rule is not activated, marking it as an invalid rule; yellow points indicate that the rule is activated too infrequently, marking it as an inefficient rule; green points indicate that although the rule is activated frequently, its activation weight is too low, marking it as an inefficient rule; purple points indicate that the rule's activation frequency and weight are sufficient to impact the model, marking it as an efficient rule.

Table 5 shows the determination of each sub-layer's rule type based on the analysis of activation weights. Comparing Table 4 with Table 5 demonstrates that the efficiency factor effectively indicates the redundancy of rules.

Using the efficiency factor for rule reduction allows inefficient and invalid rules to be excluded from optimization, significantly reducing the number of parameters involved in the optimization process. The number of parameters in the optimization process is directly related to the number of rules that must be optimized within each sub-layer. Each rule consists of 6 parameters (5 beliefs and 1 rule weight). Additionally,

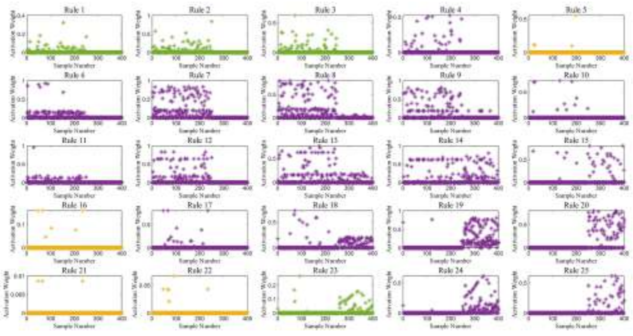

FIGURE 9. Analysis of Sub-BRB1 rule activation weights.

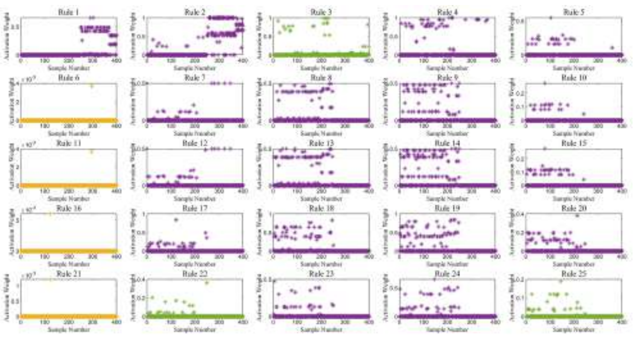

FIGURE 10. Analysis of Sub-BRB2 rule activation weights.

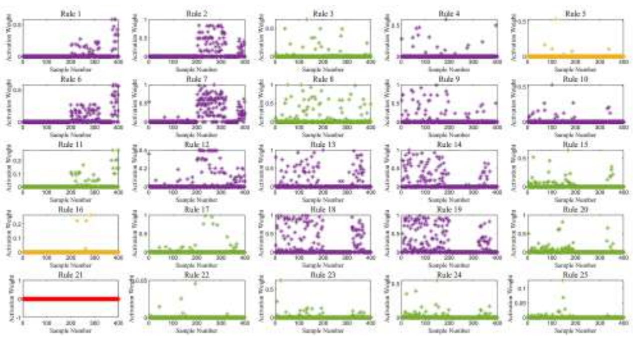

FIGURE 11. Analysis of Sub-BRB3 rule activation weights.

TABLE 5. Different types of rules in three sub-layers

Sub-layer	Invalid rule	Inefficient rule	Efficient rule
Sub-BRB1	0	Rule 1,2,3,5, 16,21,22,23	Rule 4,6,7,8,9,10,11,12,13, 14,15,17,18,19,20,24,25
Sub-BRB2	0	Rule 3,6,11, 16,21,22,25	Rule 1,2,4,5,7,8,9,10,12,13, 14,15,17,18,19,20,23,24
Sub-BRB3	Rule 21	Rule 3,5,8,11,15,16, 17,20,22,23,24,25	Rule 1,2,4,6,7,9, 10,12,13,14,18,19

each sub-layer incorporates two renal function indicators, each with its own attribute weight. Prior to rule reduction, the optimization process for each sub-layer involves 25 rules and 2 attribute weights, totaling 152 parameters. Therefore, the entire model requires the optimization of 456 parameters before rule reduction. Table 6 illustrates the optimization difficulty from the perspective of the number of parameters. After rule reduction, the number of parameters to optimize decreases significantly. Sub-BRB1 now requires only 104 parameters, Sub-BRB2 requires 120 parameters, and Sub-BRB3 requires just 74 parameters. This leads to a reduction of 168 parameters, which accounts for nearly 40% of the total parameters before rule reduction. This simplification not only reduces model complexity but also lessens the optimization workload, ultimately enhancing the efficiency and stability of the model when applied to larger datasets.

TABLE 6. Comparison of optimization parameters before and after rule reduction

Layer	Before rule reduction	After rule reduction	Comparation
Sub-BRB1	152	104	48 ↓
Sub-BRB2	152	110	42 ↓
Sub-BRB3	152	74	78 ↓
DBRB-I	456	288	168 ↓

Figure 12 compares the difficulty of optimization for the CKD diagnostic model based on DBRB-I before and after rule reduction, from the perspective of optimization time. The figure shows the time consumption in the three sub-layers before and after rule reduction. Before reduction, the times were 59.817 seconds, 60.564 seconds, and 60.704 seconds, respectively. After reduction, the times were reduced to 51.172 seconds, 52.987 seconds, and 47.355 seconds. Table 7 summarizes the consumption of optimization time after rule reduction. The result indicates that the time consumption in the three sub-layers decreased by approximately 14.4%, 12.5%, and 22.1%, respectively, and the overall model time decreased by 16.3%. The experimental results demonstrate that eliminating redundant parameters during the optimization process significantly reduces both optimization time and complexity. The rule reduction method improves the diagnostic efficiency of the CKD diagnostic model based on DBRB-I, thereby shortening decision-making time. By enhancing diagnostic efficiency, the model can respond more rapidly to clinical needs, enabling doctors to make accurate diagnoses in less time. This not only improves the efficiency of clinical workflows but also enhances the patient's treatment experience.

TABLE 7. Optimization time consumption after rule reduction

Layer	Before rule reduction	After rule reduction	Comparation
Sub-BRB1	59.817s	51.172s	14.4% ↓
Sub-BRB2	60.564s	52.987s	12.5% ↓
Sub-BRB3	60.704s	47.355s	22.1% ↓
DBRB-I	181.085s	151.514s	16.3% ↓

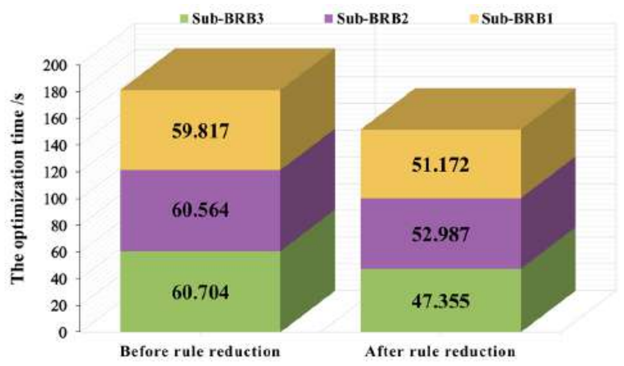

FIGURE 12. Comparison of optimization time before and after rule reduction.

2) Interpretability assessment

Expert knowledge is the source and basis for the interpretability of the CKD diagnostic model based on DBRB-I. This paper places a strong trust in expert knowledge and uses it as the initial belief distribution to construct the CKD diagnostic model based on DBRB-I. Although the optimization process may reduce the model's interpretability, meaning that the final belief distribution output by the model might deviate to some extent from the expert knowledge, the deviation should not be excessive. This indicates that the smaller the deviation between the belief distribution output of the CKD diagnostic model and the expert knowledge, the stronger the model interpretability.

Figures 13, 14, and 15 compare the belief distributions of Sub-BRB1, Sub-BRB2, and Sub-BRB3 with expert knowledge, both with and without interpretability constraints. In the figures, the yellow line represents expert knowledge, the purple line represents the DBRB without interpretability constraints, and the green line represents the DBRB-I with interpretability constraints. Rules without a yellow line, such as Rule 1, Rule 2, and Rule 3 in Figure 13, are considered invalid or inefficient rules that are not optimized. The model continues to use the initial values set by expert knowledge, so the yellow line is covered by the green line. Focus on the red circles, which highlight efficient rules where the DBRB significantly deviates from expert knowledge. The comparison shows that the green lines (DBRB-I) are closer to the yellow lines (expert knowledge) than the purple lines (DBRB without constraints) in each rule. This indicates that the interpretability of the CKD diagnostic model based on DBRB-I surpasses that of the model based on DBRB. Additionally, the CKD diagnostic model based on DBRB-I is more transparent, enabling doctors to understand the model's decision-making process, which in turn fosters greater trust in the model. This transparency not only facilitates the model's more confident application in clinical practice but also enhances its clinical value and practical feasibility, providing doctors with more reliable support.

Appendix C contains the optimized parameters without interpretability for each sub-layer of the CKD diagnostic model

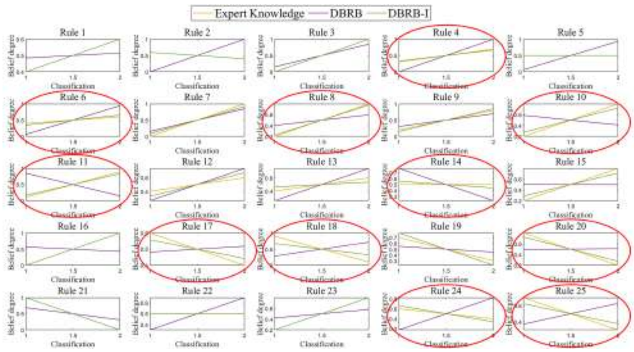

FIGURE 13. Comparison of the belief distributions of Sub-BRB1.

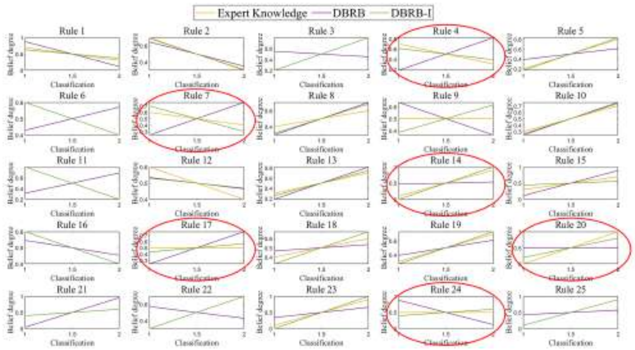

FIGURE 14. Comparison of the belief distributions of Sub-BRB2.

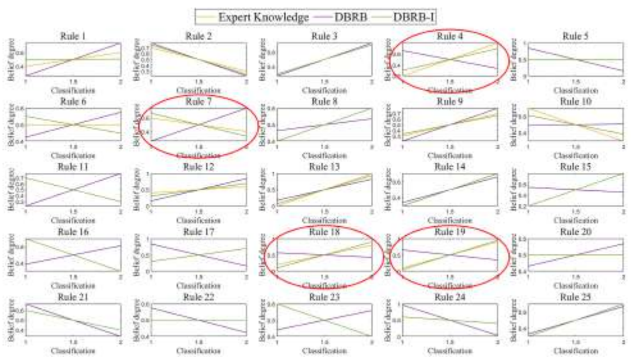

FIGURE 15. Comparison of the belief distributions of Sub-BRB3.

based on DBRB, including the optimized rule weights and the optimized belief distributions. Appendix D contains the optimized parameters with interpretability for each sub-layer of the CKD diagnostic model based on DBRB-I, including the optimized rule weights and the optimized belief distributions. The details of the optimization process for these optimized parameters can be found in Section III-C.

3) Analysis of the CKD diagnostic model based on DBRB-I

The CKD diagnostic model based on DBRB-I is constructed with three layers, each adding a new renal function indicator. Figure 16 compares the diagnostic results of Sub-BRB1, Sub-BRB2, and Sub-BRB3. Our analysis shows that the sub-layer with the additional renal function indicator diagnoses CKD more accurately than the previous sub-layer, demonstrating that the hierarchical structure of the CKD diagnostic model based on DBRB-I is meaningful in practical CKD diagnostic environments.

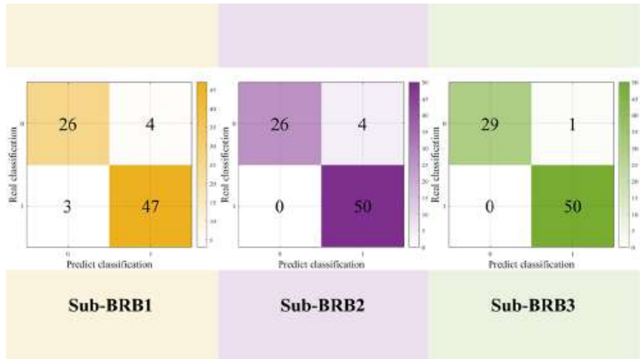

FIGURE 16. Comparison of diagnostic results in three sub-layers.

To prevent overfitting of the training samples, this paper employs 5-fold cross-validation to more comprehensively assess the performance of the CKD diagnostic model based on DBRB-I on new data. The paper divides 400 patient samples into five groups, with each group representing 20% of the total samples. In each round of the experiment, one group is used as the test set, while the remaining four groups serve as the training set. Each round consists of 20 experiments to demonstrate the robustness of the CKD diagnostic model based on DBRB-I.

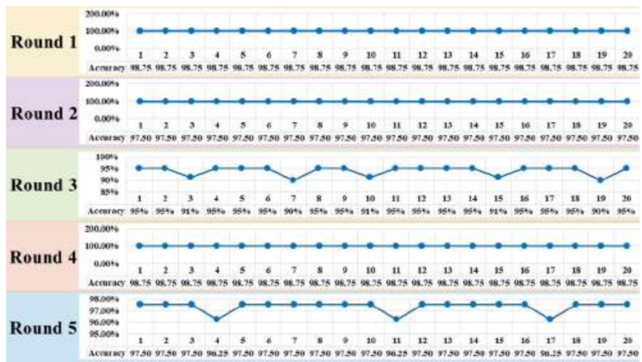

FIGURE 17. Results of 20 trials per fold in the 5-fold cross-validation experiment.

In Figure 17, it is evident that Round1, Round2, and Round4 exhibit stable diagnostic results, while Round3 and Round5 show slight fluctuations. The horizontal axis represents the number of experiments and the vertical axis represents the accuracy of each experiment. The stability in diagnostic results for Rounds 1, 2, and 4 indicates that the CKD diagnostic model based on DBRB-I performs consistently across different experimental runs for these rounds. The minor fluctuations observed in Rounds 3 and 5 suggest some variability in the model's performance, but overall the results still demonstrate the model's robustness and reliability.

4) Comparison of different CKD diagnostic models

In this paper, both horizontal and vertical comparisons were performed using the same training and testing datasets.

Horizontal Comparison: The performance of four CKD diagnostic models was compared:

1) K-Nearest Neighbors (KNN) and Support Vector Machines (SVM) are examples of machine learning methods that represent black-box models.
2) Random Forest (RF) is an example of a machine-learning method that represents a gray-box model.
3) Decision Tree (DT) is an example of a machine learning method that represents a white-box model.
4) The CKD diagnostic model based on DBRB-I.

Vertical Comparison: The performance of three CKD diagnostic models based on BRB was compared:

1) The model without deep-layer structure (BRB).
2) The model without interpretability constraints (DBRB).
3) The model that makes decisions based solely on existing expert knowledge without parameter optimization (DBRB-E).
4) The model with interpretability constraints (DBRB-I).

Figure 18 compares the diagnostic results of these eight methods using confusion matrices. A comparison of the confusion matrix reveals that the CKD diagnostic models based on DBRB and DBRB-I perform the best. However, the CKD diagnostic model based on DBRB-I offers stronger interpretability than the model based on DBRB. This indicates that the model based on DBRB-I is better suited for real-world CKD diagnostic environments, providing patients with more accurate medical support.

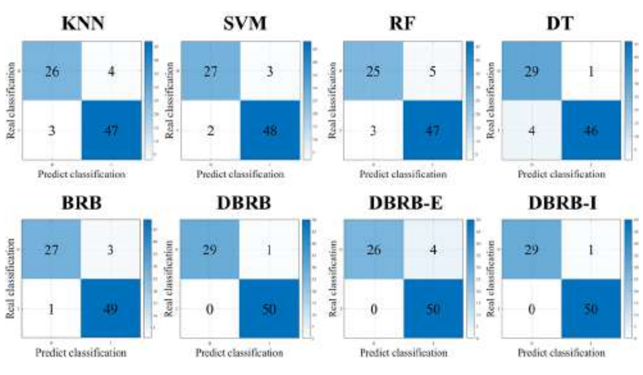

FIGURE 18. Comparison of diagnostic results for the eight methods.

V. CONCLUSION

This paper introduces a CKD diagnostic model based on DBRB-I, with the goal of enhancing diagnostic acceptance and trustworthiness. To achieve this, we initially analyzed patients' renal function indicators using the Random Forest method. By comparing the performance of varying numbers of renal function indicators within the Random Forest framework, we identified the key indicators to include in the CKD diagnostic model based on DBRB-I. These indicators were then prioritized according to their importance scores, which were calculated using the Random Forest method.

Next, we calculated efficiency factors for all rules in the initial CKD diagnostic model based on DBRB-I, which was constructed by experts, and filtered out redundant rules to simplify the optimization process. To assess whether the optimization difficulty was reduced, we evaluated it in terms of the number of parameters and time consumption. Specifically, the number of parameters to be optimized decreased by nearly 40%, and the optimization time was reduced by 16.3% after rule reduction. These results demonstrate that the proposed rule reduction method significantly decreased the model's optimization time, effectively lowering both the complexity and difficulty of the optimization process.

Finally, we incorporated constraints into the optimization algorithm to ensure that the CKD diagnostic model based on DBRB-I remains aligned with expert knowledge throughout the optimization process. These measures effectively enhance the model's reliability and practicality.

This paper presents three innovations. First, it introduces a deep BRB model into CKD diagnosis, effectively addressing the limitations of traditional BRB models regarding rule explosion and scalability. Second, it designs a new efficiency-based rule reduction method that combines expert knowledge, enhancing the transparency and reliability of rule reduction. Third, it proposes a new optimization algorithm with constraints that integrate expert knowledge to ensure that the model's interpretability is not compromised by the optimization process. The effectiveness of the model was validated through a case study on CKD diagnosis, which demonstrated that the CKD diagnostic model based on DBRB-I performs better in the practical diagnostic environment.

However, the dataset used to validate the CKD diagnostic model based on DBRB-I has a limitation. The size of the dataset has certain limitations. This dataset was chosen because it is the only one among existing open-source data without missing values and that meets CKD diagnostic requirements. Although the size of the dataset may affect the generalizability of the results, we conducted rigorous robustness tests, including cross-validation and repeated experiments, in the case study, which showed good stability. In the future, we plan to expand the dataset and further validate the research findings to enhance the results' broader applicability and reliability.

APPENDIX A THE INITIAL PARAMETERS

Rule	RW	RFI		The initial belief	
	ϖ	χ_1	χ_2	β_1	β_2
1	0.3	VL	VL	0.4	0.6

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Rule	RW	RFI		The initial belief	
	ϖ	χ_1	χ_2	β_1	β_2
2	0.4	VL	L	0.6	0.4
3	0.4	VL	M	0.0	1.0
4	0.9	VL	H	0.3	0.7
5	0.2	VL	VH	0.5	0.5
6	1.0	L	VL	0.4	0.6
7	0.9	L	L	0.0	1.0
8	0.7	L	M	0.2	0.8
9	1.0	L	H	0.2	0.8
10	0.6	L	VH	0.2	0.8
11	1.0	M	VL	0.1	0.9
12	0.9	M	L	0.4	0.6
13	0.7	M	M	0.4	0.6
14	1.0	M	H	0.5	0.5
15	1.0	M	VH	0.2	0.8
16	0.4	H	VL	0.0	1.0
17	0.8	H	L	0.6	0.4
18	0.4	H	M	0.9	0.1
19	1.0	H	H	0.7	0.3
20	1.0	H	VH	0.8	0.2
21	0.1	VH	VL	1.0	0.0
22	0.1	VH	L	0.5	0.5
23	0.1	VH	M	0.2	0.8
24	0.5	VH	H	0.6	0.4
25	0.9	VH	VH	0.7	0.3
1	1.0	VL	VL	0.7	0.3
2	1.0	VL	L	0.7	0.3
3	1.0	VL	M	0.2	0.8
4	0.2	VL	H	0.7	0.3
5	1.0	VL	VH	0.2	0.8
6	0.6	L	VL	0.6	0.4
7	1.0	L	L	0.6	0.4
8	1.0	L	M	0.4	0.6
9	1.0	L	H	0.5	0.5
10	1.0	L	VH	0.3	0.7
11	1.0	M	VL	0.8	0.2
12	0.5	M	L	0.6	0.4
13	0.7	M	M	0.3	0.7
14	1.0	M	H	0.1	0.9
15	0.2	M	VH	0.3	0.7
16	0.4	H	VL	0.6	0.4
17	1.0	H	L	0.5	0.5
18	1.0	H	M	0.4	0.6
19	1.0	H	H	0.3	0.7

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(Continued)

Rule	RW	RFI		The initial belief	
	ϖ	χ_1	χ_2	β_1	β_2
20	1.0	H	VH	0.0	1.0
21	1.0	VH	VL	0.4	0.6
22	1.0	VH	L	0.3	0.7
23	0.5	VH	M	0.1	0.9
24	0.9	VH	H	0.5	0.5
25	0.8	VH	VH	0.1	0.9
1	0.7	VL	VL	0.4	0.6
2	0.8	VL	L	0.7	0.3
3	0.4	VL	M	0.3	0.7
4	1.0	VL	H	0.2	0.8
5	0.4	VL	VH	0.5	0.5
6	0.7	L	VL	0.5	0.5
7	1.0	L	L	0.6	0.4
8	0.9	L	M	0.4	0.6
9	1.0	L	H	0.3	0.7
10	1.0	L	VH	0.7	0.3
11	0.4	M	VL	0.7	0.3
12	0.6	M	L	0.4	0.6
13	1.0	M	M	0.0	1.0
14	1.0	M	H	0.3	0.7
15	0.4	M	VH	0.2	0.8
16	0.1	H	VL	0.7	0.3
17	0.4	H	L	0.3	0.7
18	1.0	H	M	0.2	0.8
19	1.0	H	H	0.1	0.9
20	0.4	H	VH	0.5	0.5
21	0.0	VH	VL	0.6	0.4
22	0.4	VH	L	0.5	0.5
23	0.4	VH	M	0.6	0.4
24	0.4	VH	H	0.6	0.4
25	0.4	VH	VH	0.3	0.7

APPENDIX B THE EFFICIENCY FACTOR

Rule	Sub-BRB1	Sub-BRB2	Sub-BRB3
1	0.182535	5.78	3.274074
2	0.517125	8.393243	5.120588
3	0.468094	0.380859	0.375333
4	2.38125	3.491176	1.677778
5	0.95	2.364286	0.921429
6	4.65	0.05	5.414583
7	7.623529	1.8	8.652564
8	7.647842	4.171429	0.562744

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Rule	Sub-BRB1	Sub-BRB2	Sub-BRB3
9	5.26954	4.385714	3.2125
10	1.55	1.528947	1.645455
11	2.719048	0.05	0.485368
12	4.961905	2.312069	5
13	6.662712	5.359677	5.381579
14	6.200467	6.065385	5.560417
15	2.611765	2.591026	0.336019
16	0.8	0.05	0.95
17	1.622222	2.559091	0.415093
18	5.093182	3.783333	6.411062
19	8.026389	4.810256	7.926224
20	4.385294	3.2	0.416826
21	0.15	0.05	0
22	0.3	0.357333	0.013
23	0.293898	2.57973	0.218636
24	4.855747	3.086364	0.195
25	2.854878	0.296667	0.063303

APPENDIX C THE OPTIMIZED PARAMETERS WITHOUT INTERPRETABILITY

Rule	RW	RFI		The optimized belief	
	ϖ	χ_1	χ_2	β_1	β_2
1	0.177412	VL	VL	0.484654	0.515346
2	0.0	VL	L	0.0	1.0
3	0.509146	VL	M	0.152246	0.847754
4	0.29219	VL	H	0.0	1.0
5	0.337471	VL	VH	0.065012	0.934988
6	0.0	L	VL	0.0618	0.9382
7	0.333433	L	L	0.154238	0.845762
8	0.320919	L	M	0.402925	0.597075
9	0.886596	L	H	0.305397	0.694603
10	0.543651	L	VH	0.587941	0.412059
11	0.089817	M	VL	0.852462	0.147538
12	0.713013	M	L	0.262966	0.737034
13	0.750472	M	M	0.256962	0.743038
14	0.439834	M	H	0.777184	0.222816
15	0.235707	M	VH	0.502824	0.497176
16	0.382029	H	VL	0.57134	0.42866
17	0.771495	H	L	0.481474	0.518526
18	0.0	H	M	0.283799	0.716201
19	0.281316	H	H	0.550733	0.449267
20	0.218637	H	VH	0.491397	0.508603
21	0.634487	VH	VL	0.68865	0.31135

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(Continued)

Rule	RW	RFI		The optimized belief	
	ϖ	χ_1	χ_2	β_1	β_2
22	1.0	VH	L	0.350192	0.649808
23	0.275783	VH	M	0.420424	0.579576
24	0.17285	VH	H	0.17635	0.82365
25	0.623417	VH	VH	0.370873	0.629127
1	0.656352	VL	VL	0.882065	0.117935
2	0.17615	VL	L	0.647849	0.352151
3	0.62249	VL	M	0.550601	0.449399
4	0.509511	VL	H	0.173635	0.826365
5	0.352456	VL	VH	0.388279	0.611721
6	0.800171	L	VL	0.428626	0.571374
7	0.385774	L	L	0.242414	0.757586
8	0.30863	L	M	0.29523	0.70477
9	0.725404	L	H	0.635529	0.364471
10	0.66882	L	VH	0.244705	0.755295
11	0.422452	M	VL	0.311485	0.688515
12	0.199034	M	L	0.535334	0.464666
13	0.666177	M	M	0.186682	0.813318
14	0.420593	M	H	0.46253	0.53747
15	0.0	M	VH	0.110521	0.889479
16	1.0	H	VL	0.546269	0.453731
17	0.373521	H	L	0.244476	0.755524
18	0.400154	H	M	0.464611	0.535389
19	0.39506	H	H	0.388228	0.611772
20	0.650891	H	VH	0.496298	0.503702
21	0.536908	VH	VL	0.042685	0.957315
22	0.825045	VH	L	0.572553	0.427447
23	0.619996	VH	M	0.342064	0.657936
24	0.402314	VH	H	0.877784	0.122216
25	0.303382	VH	VH	0.427288	0.572712
1	0.250263	VL	VL	0.261115	0.738885
2	0.681106	VL	L	0.779624	0.220376
3	0.857169	VL	M	0.274032	0.725968
4	0.24463	VL	H	0.662645	0.337355
5	0.524554	VL	VH	0.846141	0.153859
6	0.171755	L	VL	0.424252	0.575748
7	0.965722	L	L	0.255542	0.744458
8	0.529596	L	M	0.463757	0.536243
9	0.779566	L	H	0.208768	0.791232
10	0.523316	L	VH	0.490899	0.509101
11	0.0	M	VL	0.226325	0.773675
12	0.685295	M	L	0.155684	0.844316
13	0.385829	M	M	0.176428	0.823572
14	0.409392	M	H	0.339548	0.660452

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Rule	RW	RFI		The optimized belief	
	ϖ	χ_1	χ_2	β_1	β_2
15	0.139162	M	VH	0.544351	0.455649
16	0.447975	H	VL	0.386439	0.613561
17	0.419957	H	L	0.838398	0.161602
18	0.474133	H	M	0.571127	0.428873
19	0.81202	H	H	0.655513	0.344487
20	0.943775	H	VH	0.43116	0.56884
21	0.722154	VH	VL	0.665511	0.334489
22	0.314948	VH	L	0.575634	0.424366
23	0.460693	VH	M	0.440689	0.559311
24	0.552843	VH	H	0.968255	0.031745
25	0.613271	VH	VH	0.330712	0.669288

APPENDIX D THE OPTIMIZED PARAMETERS WITH INTERPRETABILITY

Rule	RW	RFI		The optimized belief	
	ϖ	χ_1	χ_2	β_1	β_2
1	0.3	VL	VL	0.4	0.6
2	0.4	VL	L	0.6	0.4
3	0.4	VL	M	0.0	1.0
4	0.660305	VL	H	0.329409	0.670591
5	0.2	VL	VH	0.5	0.5
6	0.789695	L	VL	0.34527	0.65473
7	0.858433	L	L	0.083473	0.916527
8	0.420784	L	M	0.221503	0.778497
9	0.849879	L	H	0.149942	0.850058
10	0.450121	L	VH	0.274304	0.725696
11	0.640718	M	VL	0.166849	0.833151
12	0.718564	M	L	0.327003	0.672997
13	0.523081	M	M	0.463451	0.536549
14	0.738459	M	H	0.557058	0.442942
15	0.797356	M	VH	0.286647	0.713353
16	0.4	H	VL	0.0	1.0
17	0.602644	H	L	0.558175	0.441825
18	0.256552	H	M	0.676319	0.323681
19	0.886897	H	H	0.778751	0.221249
20	0.823067	H	VH	0.725674	0.274326
21	0.1	VH	VL	1.0	0.0
22	0.1	VH	L	0.5	0.5
23	0.1	VH	M	0.2	0.8
24	0.353867	VH	H	0.662401	0.337599
25	0.792584	VH	VH	0.624268	0.375732
1	0.724097	VL	VL	0.632493	0.367507

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Rule	RW	RFI		The optimized belief	
	ϖ	χ_1	χ_2	β_1	β_2
2	0.775903	VL	L	0.691273	0.308727
3	1.0	VL	M	0.2	0.8
4	0.141028	VL	H	0.612868	0.387132
5	0.858972	VL	VH	0.1686	0.8314
6	0.6	L	VL	0.6	0.4
7	0.740149	L	L	0.7	0.3
8	0.759851	L	M	0.318028	0.681972
9	0.750664	L	H	0.385429	0.614571
10	0.749336	L	VH	0.263257	0.736743
11	1.0	M	VL	0.8	0.2
12	0.605249	M	L	0.53095	0.46905
13	0.494751	M	M	0.255466	0.744534
14	0.687462	M	H	0.0	1.0
15	0.412538	M	VH	0.439522	0.560478
16	0.4	H	VL	0.6	0.4
17	0.567755	H	L	0.433336	0.566664
18	0.86449	H	M	0.325319	0.674681
19	1.0	H	H	0.264612	0.735388
20	1.0	H	VH	0.201403	0.798597
21	1.0	VH	VL	0.4	0.6
22	1.0	VH	L	0.3	0.7
23	0.687488	VH	M	0.0	1.0
24	0.492571	VH	H	0.400439	0.599561
25	0.8	VH	VH	0.1	0.9
1	0.546869	VL	VL	0.49386	0.50614
2	0.678126	VL	L	0.749504	0.250496
3	0.4	VL	M	0.3	0.7
4	0.575736	VL	H	0.301025	0.698975
5	0.4	VL	VH	0.5	0.5
6	0.677767	L	VL	0.551577	0.448423
7	0.757977	L	L	0.670825	0.329175
8	0.9	L	M	0.4	0.6
9	0.978619	L	H	0.334795	0.665205
10	0.685058	L	VH	0.614651	0.385349
11	0.4	M	VL	0.7	0.3
12	0.639707	M	L	0.306664	0.693336
13	0.823253	M	M	0.068261	0.931739
14	0.834599	M	H	0.297069	0.702931
15	0.4	M	VH	0.2	0.8
16	0.1	H	VL	0.7	0.3
17	0.4	H	L	0.3	0.7
18	0.896259	H	M	0.1	0.9
19	0.898925	H	H	0.048819	0.951181

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Rule	RW	RFI		The optimized belief	
	ϖ	χ_1	χ_2	β_1	β_2
20	0.4	H	VH	0.5	0.5
21	0.0	VH	VL	0.6	0.4
22	0.4	VH	L	0.5	0.5
23	0.4	VH	M	0.6	0.4
24	0.4	VH	H	0.6	0.4
25	0.4	VH	VH	0.3	0.7

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